

PAST EXAMINATION QUESTIONS (FP2)

TOPIC 14: INTEGRATION (3) – LENGTH OF ARC, AREA OF SURFACE

1. A curve C is defined parametrically by

$$x = \frac{2}{e^t + e^{-t}} \quad \text{and} \quad y = \frac{e^t - e^{-t}}{e^t + e^{-t}},$$

for $0 \leq t \leq 1$. The area of the surface generated when C is rotated through 2π radians about the x -axis is denoted by S .

(i) Show that $S = 4\pi \int_0^1 \frac{e^t - e^{-t}}{(e^t + e^{-t})^2} dt$. [5]

(ii) Using the substitution $u = e^t + e^{-t}$, or otherwise, find S in terms of π and e . $\left(4\pi \left(\frac{1}{2} - \frac{1}{e+e^{-1}}\right)\right)$ [3]

May/June 2019/11/12/5

2. The curve C is defined parametrically by

$$x = 18t - t^2 \quad \text{and} \quad y = 8t^{\frac{3}{2}},$$

where $0 < t \leq 4$.

- (i) Showing that at all points of C ,

$$\frac{d^2y}{dx^2} = \frac{3(9+t)}{2t^{\frac{1}{2}}(9-t)^3}. \quad [4]$$

- (iii) Find the area of the surface generated when C is rotated through 2π radians about the x -axis, showing full working. $\left(\frac{169984\pi}{35}\right)$ [6]

Oct/Nov 2018/12/11E (part)

3. The curve C is defined parametrically by

$$x = e^t - t, \quad y = 4e^{\frac{1}{2}t}.$$

Find the length of the arc C from the point where $t = 0$ to the point where $t = 3$. $(2 + e^3)$ [5]

May/June 2018/11/12/1

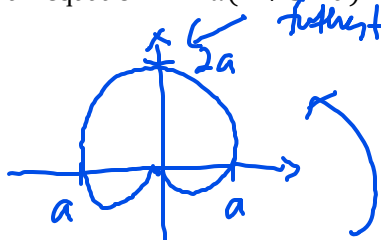
4. The polar equation of a curve C is $r = a(1 + \cos \theta)$ for $0 \leq \theta < 2\pi$, where a is a positive constant.

- (iv) Find the arc length of C between the point where $\theta = 0$ and the point where $\theta = \frac{1}{3}\pi$. $(2a)$ [5]

Oct/Nov 2017/11/12/13/11O

5. The curve C has polar equation $r = a(1 + \sin \theta)$ for $-\pi < \theta \leq \pi$, where a is a positive constant.

- (i) Sketch C . [2]



- (iii) Show that the length of the arc of C from the pole to the point furthest from the pole is given by

$$s = (\sqrt{2})a \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \sqrt{1 + \sin \theta} d\theta. \quad [3]$$

- (iv) Show that the substitution $u = 1 + \sin \theta$ reduces this integral for s to $(\sqrt{2})a \int_0^2 \frac{1}{\sqrt{(2-u)}} du$. Hence evaluate s . ($s = 4a$) [4]

May/June 2017/11/12/11 (part)

6. The curve C has equation $y = \frac{1}{2}(e^x + e^{-x})$ for $0 \leq x \leq 4$.

- (ii) Show that $\frac{ds}{dx} = \frac{1}{2}(e^x + e^{-x})$, where s denotes the arc length of C , and find the surface area generated when C is rotated through 2π radians about the x -axis. $\left(\frac{\pi}{2}\left(\frac{e^8}{2} + 8 - \frac{e^{-8}}{2}\right)\right)$ [4]

May/June 2017/11/12/12E

7. A curve C has parametric equations

$$x = \frac{2}{5}t^{\frac{5}{2}} - 2t^{\frac{1}{2}}, \quad y = \frac{4}{3}t^{\frac{3}{2}}, \quad \text{for } 1 \leq t \leq 4.$$

- (i) Find the exact value of the arc length of C . [5]
(ii) Find also the exact value of the surface area generated when C is rotated through 2π radians about the x -axis. [3]

May/June 2017/13/5

8. A curve C has parametric equations

$$x = 1 - 3t^2, \quad y = t(1 - 3t^2), \quad \text{for } 0 \leq t \leq \frac{1}{\sqrt{3}}.$$

Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (1 + 9t^2)^2$. [2]

Hence find

- (i) the arc length of C . ($s = \frac{2}{\sqrt{3}}$) [2]
(ii) The surface area generated when C is rotated through 2π radians about the x -axis. $\left(\frac{4\sqrt{3}}{45}\right)$ [3]

Oct/Nov 2016/11/12/13/11O

9. A curve C has parametric equations

$$x = e^{2t} \cos 2t, \quad y = e^{2t} \sin 2t, \quad \text{for } -\frac{1}{2}\pi \leq t \leq \frac{1}{2}\pi.$$

Find the arc length of C . ($2\sqrt{2}(e^\pi - e^{-\pi})$ or $2\sqrt{2} \sinh \pi$ or 32.7) [6]

Find the area of the surface generated when C is rotated through 2π radians about the x -axis. [8]

$$\left(\frac{2\sqrt{2}\pi}{5}(e^{2\pi} - e^{-2\pi})\right) \text{ or } \frac{4\sqrt{2}\pi}{5} \sinh 2\pi \text{ or } 952$$

May/June 2016/11/12/11E

10. The curve C has equation $y = -\ln(1 - x^2)$ for $-\frac{1}{2} \leq x \leq \frac{1}{2}$. Show that

$$1 + \left(\frac{dy}{dx}\right)^2 = \left(\frac{1 + x^2}{1 - x^2}\right)^2. \quad [2]$$

Show further that $\frac{1+x^2}{1-x^2}$ may be expressed in the form $\frac{P}{1+x} + \frac{Q}{1-x} + R$, where P, Q and R are constants to be determined. ($P = 1, Q = 1, R = -1$) [2]

Find the exact arc length of C . ($2 \ln 3 - 1$) [4]

May/June 2016/13/4

11. The curve C has equation $y = \ln\left(\tanh \frac{1}{2}x\right)$, for $x > 0$.

(i) Show that $\frac{dy}{dx} = \operatorname{cosech} x$. [3]

(ii) For positive integers n , the length of the arc of C between $x = n$ and $x = 2n$ is L_n .

(a) Show by calculus that, when n is large, $L_n \approx n$. [5]

(b) Explain how this result corresponds to the shape of C . [2]

CAIE 9795/May/June 2016/1/12

12. The curve C has polar equation $r = a(1 - \cos \theta)$ for $0 \leq \theta < 2\pi$. Sketch C . [2]

Find the area of the region enclosed by the arc of C for which $\frac{1}{2}\pi \leq \theta \leq \frac{3}{2}\pi$, the half-line $\theta = \frac{1}{2}\pi$ and the half-line $\theta = \frac{3}{2}\pi$. $\left(a^2 \left(\frac{3}{4}\pi + 2\right)\right)$ [5]

Show that

$$\left(\frac{ds}{d\theta}\right)^2 = 4a^2 \sin^2\left(\frac{1}{2}\theta\right),$$

where s denotes arc length, and find the length of the arc of C for which $\frac{1}{2}\pi \leq \theta \leq \frac{3}{2}\pi$. [7]

Oct/Nov 2015/11/12/13/11O

13. The curve C has parametric equations

$$x = 4t + 2t^{\frac{3}{2}}, y = 4t - 2t^{\frac{3}{2}}, \text{ for } 0 \leq t \leq 4.$$

Find the arc length of C , giving your answer correct to 3 significant figures. (32.6) [6]

May/June 2015/11/12/9

14. The curve C has polar equation $r = e^{4\theta}$ for $0 \leq \theta \leq \alpha$, where α is measured in radians. The length of C is 2015. Find the value of α . (1.89) [6]

May/June 2015/13/2

15. A curve C has parametric equations

$$x = e^t \cos t, \quad y = e^t \sin t, \quad \text{for } 0 \leq t \leq \frac{1}{2}\pi.$$

Find the arc length of C . $\left(\sqrt{2}\left(e^{\frac{\pi}{2}} - 1\right)\right)$ [6]

Oct/Nov 2014/11/12/13/2

16. The curve C has parametric equations

$$x = t^2, \quad y = t - \frac{1}{3}t^3, \quad \text{for } 0 \leq t \leq 1.$$

Find

- (i) The arc length of C . $\left(\frac{4}{3}\right)$ [5]
 (ii) The surface area generated when C is rotated through 2π radians about the x -axis. $\left(\frac{11\pi}{9}\right)$ [5]

May/June 2014/11/12/8

17. (i) (a) Show that $\tanh x = \frac{e^{2x}-1}{e^{2x}+1}$. [2]
 (b) Hence, or otherwise, show that, if $\tanh x = \frac{1}{k}$ for $k > 1$, then $x = \frac{1}{2}\ln\left(\frac{k+1}{k-1}\right)$ and find an expression in terms of k for $\sinh 2x$. $\left(\frac{2k}{k^2-1}\right)$ [4]
 (ii) A curve has equation $y = \frac{1}{2}\ln(\tanh x)$ for $\alpha \leq x \leq \beta$, where $\tanh \alpha = \frac{1}{3}$ and $\tanh \beta = \frac{1}{2}$. Find, in its simplest exact form, the arc length of this curve. $\left(\ln \frac{4}{3}\right)$ [10]

CAIE 9795/May/June 2014/1/12

18. The curve C has parametric equations

$$x = e^t - 4t + 3, \quad y = 8e^{\frac{1}{2}t}, \quad \text{for } 0 \leq t \leq 2.$$

- (i) Find, in terms of e , the length of C . $(e^2 + 7)$ [5]
 (ii) Find, in terms of π and e , the area of the surface generated when C is rotated through 2π radians about the x -axis. $\left(16\pi\left(\frac{2}{3}e^3 + 8e - \frac{26}{3}\right)\right)$ [5]

May/June 2014/13/6

19. The curve C has polar equation $r = 2e^\theta$, for $\frac{1}{6}\pi \leq \theta \leq \frac{1}{2}\pi$. Find

- (ii) the length of C . $\left(2\sqrt{2}\left[e^{\frac{\pi}{2}} - e^{\frac{\pi}{6}}\right]\right)$ [3]

Oct/Nov 2013/11/12/1 (part)

20. (a) Let $I_n = \int_0^\alpha \cosh^n x \, dx$ for integers $n \geq 0$, where $\alpha = \ln 2$.

(i) Prove that, for $n \geq 2$, $nI_n = \frac{3 \times 5^{n-1}}{4^n} + \frac{1}{4}(n-1)I_{n-2}$. [5]

(ii) A curve has parametric equations $x = 12 \sinh t + 4 \sinh^3 t$, $y = 3 \cosh^4 t$, $0 \leq t \leq \ln 2$. Find the length of the arc of this curve, giving your answer in the form $a + b \ln 2$ for rational numbers a and b . ($a = \frac{2205}{256}$, $b = \frac{9}{2}$) [8]

CAIE 9795/Oct/Nov 2013/1/13 (part)

21. The curve C has parametric equations

$$x = t^2, \quad y = t - \frac{1}{3}t^3, \quad \text{for } 0 \leq t \leq 1.$$

Find the surface area generated when C is rotated through 2π radians about x -axis. ($\frac{11\pi}{9}$) [6]
Oct/Nov 2013/11/12/9

22. [In this question you may use, without proof, the formula $\int \sec x \, dx = \ln(\sec x + \tan x) + c$]

(b) The curve C has equation $y = -\ln(\cos x)$, for $0 \leq x \leq \frac{1}{3}\pi$. Find the arc length of C . ($\ln(2 + \sqrt{3})$) [4]

Oct/Nov 2013/13/6

23. The curve C has equation $y = 2 \sec x$, for $0 \leq x \leq \frac{1}{4}\pi$. Show that the arc length s of C is given by

$$s = \int_0^{\frac{1}{4}\pi} (2 \sec^2 x - 1) dx. \quad [4]$$

Find the exact value of s . ($s = 2 - \frac{1}{4}\pi$) [2]

The surface area generated when C is rotated through 2π radians about the x -axis is denoted by S . Show that

(i) $S = 4\pi \int_0^{\frac{1}{4}\pi} (2 \sec^3 x - \sec x) dx,$ [3]

(ii) $\frac{d}{dx}(\sec x \tan x) = 2 \sec^3 x - \sec x.$ [3]

Hence find the exact value of S . ($S = 4\sqrt{2}\pi$) [2]

May/June 2013/11/12/11E

24. The curve C has parametric equations $x = \frac{3}{2}t^2$, $y = t^3$, for $0 \leq t \leq 2$. Find the arc length of C . ($5\sqrt{5} - 1$) [4]

May/June 2013/13/8

25. The curve C has parametric equation

$$x = \frac{1}{3}t^3 - \ln t, \quad y = \frac{4}{3}t^{\frac{3}{2}},$$

for $1 \leq t \leq 3$. Find the arc length of C . $\left(\frac{26}{3} + \ln 3\right)$ [6]

Find also the area of the surface generated when C is rotated through 2π radians about the x -axis. $\left(\pi\left(\frac{160\sqrt{3}}{3} - \frac{64}{27}\right)\right)$ [4]

Oct/Nov 2012/11/12/8

26. The curve C has parametric equations

$$x = t^2, \quad y = \frac{1}{4}t^4 - \ln t,$$

for $1 \leq t \leq 2$. Find the area of the surface generated when C is rotated through 2π radians about the y -axis. (24π) [7]

Oct/Nov 2012/13/6

27. The curve C has equation

$$y = 2\left(\frac{x}{3}\right)^{\frac{3}{2}},$$

where $0 \leq x \leq 3$. Show that the arc length of C is $2(2\sqrt{2} - 1)$. [4]

May/June 2012/11/12/10

28. The curve C has equation $y = \frac{1}{2}(e^x + e^{-x})$ for $0 \leq x \leq \ln 5$. Find

(ii) The arc length of C . $\left(\frac{12}{5}\right)$ [4]

(iii) The surface area generated when C is rotated through 2π radians about x -axis. $\left(\pi\left(\frac{156}{25} + \ln 5\right)\right)$ [4]

Oct/Nov 2011/11/12/9 (part)

29. The curve C has equation $y = \frac{1}{3}x^{\frac{1}{2}}(3 - x)$.

Show that

$$\frac{ds}{dx} = \frac{1}{2}\left(x^{-\frac{1}{2}} + x^{\frac{1}{2}}\right),$$

where s denotes arc length, and find the arc length of C . [6]

Find the area of the surface generated when C is rotated through 2π radians about the x -axis. (3π) [4]

Oct/Nov 2011/13/11E

30. The curve C has equation $y = x^{\frac{3}{2}}$.

Show that the length of the arc C from the point where $x = 5$ to the point where $x = 28$ is 139. [5]

May/June 2011/11/12/9

31. A curve C has parametric equations $x = e^t \cos t, y = e^t \sin t$, for $0 \leq t \leq \pi$. Find the arc length of C . $(\sqrt{2}(e^\pi - 1))$ [4]

Find the area of the surface generated when C is rotated through 2π radians about the x -axis. $(\frac{2\sqrt{2}\pi}{5}(e^{2\pi} + 1))$ [7]

May/June 2011/13/7

32. The curve C has equation $y = \frac{1}{4}(e^{2x} + e^{-2x})$. Show that the length of the arc C from the point where $x = 0$ to the point where $x = \frac{1}{2}$ is $\frac{e^2 - 1}{4e}$. [4]

Oct/Nov 2010/1/1

33. At any point (x, y) on the curve C ,

$$\frac{dx}{dt} = t\sqrt{t^2 + 4} \quad \text{and} \quad \frac{dy}{dt} = -t\sqrt{4 - t^2},$$

where the parameter t is such that $0 \leq t \leq 2$. Show that the length of C is $4\sqrt{2}$. [3]

Given that $y = 0$ when $t = 2$, determine the area of the surface generated when C is rotated through one complete revolution about the x -axis, leaving your answer in an exact form. $(\frac{128\sqrt{2}\pi}{15})$ [4]

May/June 2010/11/12/3

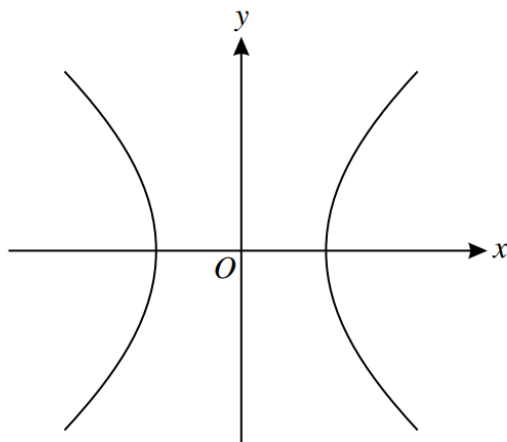
34. The parametric equations of a curve are

$$x = \cos t + t \sin t, \quad y = \sin t - t \cos t.$$

The arc of the curve joining the point where $t = 0$ to the point where $t = \frac{1}{2}\pi$ is rotated about the x -axis through one complete revolution. Find the area of the surface generated, leaving your result in terms of π . $(\frac{\pi(12 - \pi^2)}{2})$ [7]

May/June 2010/13/4

35. (ii) The diagrams shows a sketch of the hyperbola H with equation $\frac{x^2}{4} - \frac{y^2}{16} = 1$.



- (a) Find the coordinates of the points where H crosses the x -axis. $(\pm 2, 0)$ [1]
 (b) The curve J has parametric equations $x = 2 \cosh \theta$, $y = 4 \sinh \theta$, for $\theta \geq 0$. Show that these parametric equations satisfy the cartesian equation of H , and indicate on a copy of the above diagram which part of H is J . [3]
 (c) The arc of the curve J between the points where $x = 2$ and $x = 34$ is rotated once completely about the x -axis to form a surface of revolution with area S . Show that

$$S = 16\pi \int_{\alpha}^{\beta} \sinh \theta \sqrt{5 \cosh^2 \theta - 1} d\theta$$

- for suitable constants α and β . [4]
 (d) Use the substitution $u^2 = 5 \cosh^2 \theta - 1$ to show that

$$S = \frac{8\pi}{\sqrt{5}} (644\sqrt{5} - \ln(9 + 4\sqrt{5})).$$
 [9]

CAIE 9795/May/June 2010/1/12 (part)

36. (a) The curve C_1 has equation $y = -\ln(\cos x)$. Show that the length of arc of C_1 from the point where $x = 0$ to the point where $x = \frac{1}{3}\pi$ is $\ln(2 + \sqrt{3})$. [5]
 (b) The curve C_2 had equation $y = 2\sqrt{x + 3}$. The arc of C_2 joining the point where $x = 0$ to the point where $x = 1$ is rotated through one complete revolution about the x -axis. Show that the area of the surface generated is

$$\frac{8}{3}\pi(5\sqrt{5} - 8).$$
 [5]

Oct/Nov 2009/1/8

37. A curve has equation

$$y = \frac{1}{3}x^3 + 1.$$

The length of the arc of the curve joining the point where $x = 0$ to the point where $x = 1$ is denoted by s . Show that

$$s = \int_0^1 \sqrt{(1+x^4)} dx. \quad [2]$$

The surface area generated when this arc is rotated through one complete revolution about the x -axis is denoted by s . Show that

$$S = \frac{1}{9}\pi(18s + 2\sqrt{2} - 1). \quad [4]$$

[Do not attempt to evaluate s or S .]

May/June 2009/1/4

- 38.** The curve C is defined parametrically by

$$x = t^4 - 4 \ln t, \quad y = 4t^2.$$

Show that the length of the arc of C from the point where $t = 2$ to the point where $t = 4$ is

$$240 + 4 \ln 2. \quad [5]$$

Oct/Nov 2008/1/1

- 39. (ii)** A curve C in the $x - y$ plane is defined parametrically in terms of t . It is given that

$$\frac{dx}{dt} = t^4(1 - \cos 2t) \quad \text{and} \quad \frac{dy}{dt} = t^4 \sin 2t.$$

Find the length of the arc C from the point where $t = 0$ to the point where $t = \frac{1}{2}\pi$.
 $(\pi^3 - 24\pi + 48)$ [5]

May/June 2008/1/8 (part)

- 40.** A curve is defined parametrically by

$$x = at^2, \quad y = at,$$

where a is a positive constant. The part of the curve joining the point where $t = 0$ to the point where $t = \sqrt{2}$ is rotated through one complete revolution about the x -axis. Show that the area of the surface obtained is $\frac{13}{3}\pi a^2$. [4]

Oct/Nov 2007/1/1

- 41.** The curve C is defined parametrically by

$$x = t - \ln t, \quad y = 4t^{\frac{1}{2}},$$

where $t > 0$. The arc of C joining the point where $t = 1$ to the point where $t = 4$ is rotated through one complete revolution about the x -axis. Show that the area of the surface generated is $\frac{160}{3}\pi$. [6]

May/June 2007/1/2

42. The curve C has equation

$$y = x^{\frac{1}{2}} - \frac{1}{3}x^{\frac{3}{2}} + \lambda,$$

where $\lambda > 0$ and $0 \leq x \leq 3$. The length of C is denoted by s . Prove that $s = 2\sqrt{3}$. [4]

The area of the surface generated when C is rotated through one revolution about the x -axis is denoted by S . Find S in terms of λ . ($S = (4\sqrt{3}\lambda + 3)\pi$) [5]

Oct/Nov 2006/1/120

43. (a) The equation of the curve is

$$S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y = \frac{2\sqrt{3}}{3}x^{\frac{3}{2}}.$$

Find the length of the arc of the curve from the origin to the point where $x = 1$. (14) [4]

May/June 2006/1/6 (part)

44. A curve is defined parametrically by

$$x = t - 8t^{\frac{1}{2}} \quad \text{and} \quad y = \frac{16}{3}t^{\frac{3}{4}}.$$

The arc of this curve joining the point where $t = 1$ to the point where $t = 4$ is denoted by C .

- (i) Show that the length of C is 11. [5]
 (ii) Find, correct to 3 significant figures, the area of the surface generated when C is rotated through one complete revolution about the x -axis. [3]

May/June 2005/1/5

45. The curve C is defined parametrically by

$$x = a \cos^3 t, \quad y = a \sin^3 t, \quad 0 \leq t \leq \frac{1}{2}\pi,$$

where a is a positive constant. Find the area of the surface generated when C is rotated through one complete revolution about the x -axis. [5]

Oct/Nov 2004/1/2

46. The curve C has polar equation

$$r = e^{\frac{1}{5}\theta}, \quad 0 \leq \theta \leq \frac{3}{2}\pi.$$

- (ii) Find the length of C , correct to 3 significant figures. (7.99) [4]

Oct/Nov 2004/1/4 (part)

47. The curve C is defined parametrically by

$$x = t^3 - 3t, \quad y = 3t^2 + 1,$$

where $t > 1$.

- (i) Show that $\frac{d^2y}{dx^2}$ is negative at every point of C . [5]
 (ii) The arc of C joining the point where $t = 2$ to the point where $t = 3$ is rotated through one complete revolution about the x -axis. Find the area of the surface generated. $\left(\frac{4588\pi}{5}\right)$ [5]

May/June 2004/1/8

48. The arc of the curve with equation

$$y = \frac{3}{8}x^{\frac{4}{3}} - \frac{3}{4}x^{\frac{2}{3}},$$

from the point where $x = 1$ to the point where $x = 8$, is denoted by C .

- (i) Find the length of C . $\left(\frac{63}{8}\right)$ [5]
 (ii) Find the area of the surface generated when C is rotated through one revolution about the y -axis. $\left(\frac{2556\pi}{35}\right)$ [3]

May/June 2002/1/8

49. (b) For the integral

$$I_n = \int_0^1 (1 + t^2)^{n+\frac{1}{2}} dt,$$

where n is any integer, you are given that

$$(2n + 2)I_n = (2n + 1)I_{n-1} + 2^{n+\frac{1}{2}}.$$

The curve C is defined parametrically by

$$x = 6t^2 + 3t^4, \quad y = 12t + 4t^3.$$

Show that the length of the arc of C between the points where $t = 0$ and $t = 1$ is

$$\frac{1}{2}\{21\sqrt{2} + 9\ln(1 + \sqrt{2})\}. \quad [7]$$

Dec 1999/1/4 (part)

50. (a) Find the length of the curve whose equation is

$$y = \cosh(x + 2)$$

from the point where $x = 0$ to the point where $x = 1$, giving your answer correct to three significant figures. (6.39)

[5]

Nov 1997/1/4 (part)

51. Starting with the definition of $\sinh$ in terms of exponentials, prove that

$$\frac{dy}{dx} = 2\left(\frac{1}{2}x\right) = x$$

$$s = \int_0^1 \sqrt{1+x^2} dx \quad \text{let } x = \sinh u$$

$$\sinh^{-1} x = \ln \{x + \sqrt{1+x^2}\}. \quad [4]$$

Show that the length of the arc of the curve $y = \frac{1}{2}x^2$, from the origin to the point where $x = 2$ is $\sqrt{5} + \frac{1}{2}\ln(2 + \sqrt{5})$. [8]

Jun 1996/1/15

52. (b) (i) Verify that

$$\frac{d}{dx} [\sinh^{-1} x + x\sqrt{x^2 + 1}] = 2\sqrt{x^2 + 1}. \quad [2]$$

(ii) Hence show that the length of the arc of the curve whose polar equation is $r = 3\theta$, from the point where $\theta = 0$ to the point where $\theta = 5$, is

$$\frac{3}{2}\ln(5 + \sqrt{26}) + \frac{15}{2}\sqrt{26}. \quad [4]$$

June 1995/1/4 (part)

53. [In this question the results

$$\int \sinh^2 u \, du = \frac{1}{4} \sinh 2u - \frac{1}{2}u + \text{constant},$$

$$\int \cosh^2 u \, du = \frac{1}{4} \sinh 2u + \frac{1}{2}u + \text{constant},$$

may be assumed without proof.]

The curve Γ has equation

$$y = \sqrt{x^2 - 1} + \cosh^{-1} x,$$

where $x \geq 1$ and $y \geq 0$.

(ii) Show that the length, s , of the arc of Γ joining the point where $x = 4$ to the point where $x = 36$ is given by

$$s = \int_4^{36} \sqrt{\left(\frac{2x}{x-1}\right)} dx. \quad [4]$$

By using the substitution $x = \cosh^2 u$, show that

$$s = \sqrt{2} \ln \left(\frac{6 + \sqrt{35}}{2 + \sqrt{3}} \right) + 6\sqrt{70} - 2\sqrt{6}. \quad [4]$$

Nov 1994/1/4 (part)

54. (a) The curve C is defined parametrically by

$$x = \tanh t, \quad y = \tanh^2 t \quad (t \in \mathbb{R}).$$

The part of C joining the origin to the point where $t = \ln 2$ is rotated about the x -axis. Show that the area, S of the surface generated is given by

$$S = 2\pi \int_0^{\ln 2} (1 + \operatorname{sech}^2 t)^{\frac{1}{2}} \sinh t \, dt. \quad [4]$$

Nov 1993/1/4 (part)

55. (a) The curve C is defined parametrically by

$$x = 2t^2, \quad y = t^4.$$

Show that the length, s , of the part of C joining the origin to the point where $t = 1$ is given by

$$s = 4 \int_0^1 t \sqrt{1 + t^4} \, dt. \quad [3]$$

By using the substitution $t^2 = \sinh u$, or otherwise, show that $s = \sqrt{2} + \ln(1 + \sqrt{2})$. [5]

June 1993/1/4 (part)

56. A curve is defined by the equation

$$y = 2 + \cosh x,$$

where $x \in \mathbb{R}$. Calculate, correct to three significant figures,

- (iii) The area of the surface generated when the arc of the curve from the point where $x = 0$ to the point where $x = 3$ is rotated through one revolution about the x -axis. (452) [6]

Nov 1992/1/4 (part)

57. (a) The arc of the curve $y = 2\sqrt{x+1}$, from the point where $x = 0$ to the point where $x = 1$, is denoted by C . Show that the area of the surface generated when C is rotated through one revolution about the x -axis is

$$\frac{8\pi}{3}(3\sqrt{3} - 2\sqrt{2}). \quad [6]$$

June 1992/1/4 (part)

58. In the same diagram, draw sketch graphs of the curve C_1 whose equation is $y = \cosh x$ and of the curve C_2 whose equation is $y = 2 \operatorname{sech} x$. [2]

- (i) Show that the x -coordinates of the points of intersection of C_1 and C_2 are $\pm \ln(1 + \sqrt{2})$. [4]

- (ii) Show that

$$\frac{d}{dx}[\tan^{-1}(e^x)] = \frac{1}{2} \operatorname{sech} x. \quad [2]$$

- (iii) Hence, or otherwise, show that the area of the finite region between the curves C_1 and C_2 is $8 \tan^{-1}(1 + \sqrt{2}) - 2\pi - 2$. [6]

June 1991/1/3

59. The curve C has equation $y = \ln \tanh \frac{x}{2}$, where $x > 0$.

- (i) Show that $\frac{dy}{dx} = \operatorname{cosech} x$. [4]
 (ii) Show that $\frac{d^2y}{dx^2}$ is negative at every point of C . [2]
 (iii) Draw a sketch of C . [2]
 (iv) Show that the length of the part of C from the point where $x = 1$ to the point where $x = 3$ is

$$\ln(1 + e^2 + e^4) - 2. \quad [6]$$

June 1990/1/4

60. Show that for the curve which is defined parametrically by $x = \sqrt{t^2 + 1}$ and $y = \sinh^{-1} t$, ($t \in -\mathbb{R}$),

$$\frac{d^2y}{dx^2} = -\frac{\sqrt{t^2 + 1}}{t^3}.$$

Find the length of the arc of the curve from the point A , where $t = 0$, to the point B , where $t = 2$. (2) [4]

Show that the area of the surface generated when the arc AB is rotated through one revolution about the x -axis is $4\pi \ln(2 + \sqrt{5}) - 2\pi(\sqrt{5} - 1)$. [5]

December 1988/1/3

61. (a) Sketch the curve $y = \coth x$ and hence show that if $z = x \operatorname{sech} x$ then $\frac{dz}{dx} = 0$ for exactly two values of x . [6]
 (b) Show that the length of the arc of the curve $y = \cosh^{-1} x$, where $y \geq 0$, from the point A where $x = 2$ to the point B where $x = 3$ is equal to $\sqrt{8} - \sqrt{3}$. [4]

Show also that the area of the surface generated when the arc AB is rotated through one revolution about the x -axis is equal to $2\pi(2\sqrt{2} \cosh^{-1} 3 - \sqrt{3} \cosh^{-1} 2 - 1)$. [4]

December 1987/1/3

62. (a) Sketch, in a single diagram, the curves $y = \sinh x$ and $y = \frac{1}{2} \cosh x$.

Show that the area of the finite region bounded by these two curves and the y -axis is $1 - \frac{1}{2}\sqrt{3}$. [6]

December 1986/1/3 (part)

63. (a) Given that $y = \tanh^{-1} \sqrt{1 - \frac{1}{x^2}}$, where $x > 1$, find $\frac{dy}{dx}$ in terms of x , simplifying your answer. $\left(\frac{1}{\sqrt{x^2-1}}\right)$ [3]

- (b) Find the length of the part of the curve $y = \cosh x$ between the point A , where $x = 0$, and the point B , where $x = \ln 2$. $\left(\frac{3}{4}\right)$ [5]

This arc AB is rotated through one revolution about the x -axis. Show that the area of the curved surface generated is $\pi \left(\frac{15}{16} + \ln 2\right)$. [6]

June 1986/1/3

64. (b) The arc of the curve $y = a \cosh \left(\frac{x}{a}\right)$, $(0 \leq x \leq a)$ is rotated through 2π about the x -axis. Prove that the area of the curved surface generated is $\frac{1}{4}\pi a^2(e^2 + 4 - e^{-2})$. [7]

December 1984/1/17 (part)

65. For the curve with equation $y = \sinh^2 x$, prove that the length of the arc of the curve between $x = 0$ and $x = a$ is $\frac{1}{2} \sinh 2a$.

This arc of the curve is rotated through 2π radians about the x -axis. Find the area of the curved surface generated. $\left(\frac{1}{8}\pi(\sinh 4a + 4a - 4 \sinh 2a)\right)$

June 1983/2/16

66. (b) The parametric equations of a curve are $x = a \cosh t$, $y = a \sinh t$, where a is a positive constant. Prove that the area of the region in the first quadrant bounded by the curve, the x -axis between $x = a$ and $x = 2a$, and the line $x = 2a$ may be expressed as

$$\frac{1}{2}a^2 \int_0^k (\cosh 2t - 1) dt,$$

where $\cosh k = 2$, and hence show that this area is approximately $1.07a^2$.

MB June 1982/1/20 (part)

67. (a) For the curve whose equation is $y = x^2$, the length of arc between the ordinates $x = 0$ and $x = \frac{1}{2} \sinh 2$, giving your answer in hyperbolic form. $\left(\frac{1}{2} + \frac{1}{8} \sinh 4\right)$

November 1979/1/16 (part)

68. Find the length of the curve $y = \cosh x$ between $x = 0$ and $x = 2$. (3.6)

Find also the area of the surface generated when this part of curve is rotated through 2π about the x -axis. (49)

Give both your answers correct to two significant figures.

June 1977/1/16 (part)

69. Sketch the graph of the function f defined by $f: x \rightarrow \tanh^{-1} x$, $-1 < x < 1$.

Find the area of the finite region in the first quadrant between the above curve, the x -axis and the line $x = \frac{1}{2}$. (You need not obtain the numerical value of your result.)

Show that $f(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, $-1 < x < 1$.

November 1976/1/16

70. (b) Sketch the curve represented in cartesian coordinates by the equation

$$y = \cosh x.$$

The arc of this curve for which $0 \leq x \leq 2$ is rotated through four right angles about the x -axis. Find the area of the curved surface of revolution is generated. **(49)**

June 1975/1/16

71. Prove that $\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$, where $|x| < 1$, and differentiate this function.

Evaluate, to 3 significant figures, the area bounded by the curve $y = \tanh^{-1} x$, the x -axis and the line $x = \frac{1}{2}$. **(0.131)**

November 1973/1/15

72. Prove, from the definition of the sinh function, that $\sinh^{-1} x = \ln\{x + \sqrt{1+x^2}\}$.

By using the substitution $x = \frac{1}{2} \sinh u$, or some other substitution, in an appropriate integral, prove that the length of the curve $y = x^2$ between the points $\left(-\frac{20}{9}, \frac{400}{81}\right)$ and $\left(\frac{20}{9}, \frac{400}{81}\right)$ is $\ln 3 + \frac{820}{81}$.

June 1972/1/15